

Embedded Systems & IoT

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Practical Exercise 2: Circuit Design — Part A Calculation

Circuit A — Series Circuit

Supply Voltage: 125 V | R1 = 20 Ω, R2 = 30 Ω, R3 = 50 Ω

In a series circuit, the same current flows through all resistors. Voltage drops add up to the supply voltage.

1. Total Resistance

$$R_{\text{total}} = R_1 + R_2 + R_3 = 20 + 30 + 50$$

$$R_{\text{total}} = 100 \, \Omega$$

2. Total Current from Power Supply

$$I = V / R = 125 / 100$$

$$I_{\text{total}} = 1.25 \, \text{A}$$

3. Current Through Each Resistor

In a series circuit, current is the same through all components:

$$I_{R1} = I_{R2} = I_{R3} = 1.25 \, \text{A}$$

4. Voltage Drop Across Each Resistor

$$V = I \times R$$

5. Power Dissipated Through Each Resistor

$$P = V \times I$$

Resistor	Resistance (Ω)	Current (A)	Voltage Drop (V)	Power Dissipated (W)
R1	20	1.25	$1.25 \times 20 = 25.00 \, \text{V}$	$25.00 \times 1.25 = 31.25 \, \text{W}$

Resistor	Resistance (Ω)	Current (A)	Voltage Drop (V)	Power Dissipated (W)
R2	30	1.25	$1.25 \times 30 = 37.50 \text{ V}$	$37.50 \times 1.25 = 46.875 \text{ W}$
R3	50	1.25	$1.25 \times 50 = 62.50 \text{ V}$	$62.50 \times 1.25 = 78.125 \text{ W}$
Total	100 Ω	1.25 A	125.00 V	156.25 W

Circuit B — Parallel Circuit

Supply Voltage: 125 V | R1 = 20 Ω, R2 = 100 Ω, R3 = 50 Ω

In a parallel circuit, voltage across each resistor equals the supply voltage. Total current is the sum of branch currents.

1. Total Resistance

$$1/R_{\text{total}} = 1/R1 + 1/R2 + 1/R3 = 1/20 + 1/100 + 1/50$$

$$= 5/100 + 1/100 + 2/100 = 8/100$$

$$R_{\text{total}} = 100/8$$

$$R_{\text{total}} = 12.5 \text{ Ω}$$

2. Total Current from Power Supply

$$I = V / R = 125 / 12.5$$

$$I_{\text{total}} = 10 \text{ A}$$

3. Current Through Each Resistor

$$I = V / R \quad (V = 125 \text{ V for all branches})$$

4. Voltage Drop Across Each Resistor

In a parallel circuit, each resistor is directly across the supply:

$$V_{R1} = V_{R2} = V_{R3} = 125 \text{ V}$$

5. Power Dissipated Through Each Resistor

$P = V \times I$

Resistor	Resistance (Ω)	Voltage Drop (V)	Current (A)	Power Dissipated (W)
R1	20	125	125/20 = 6.25 A	125 × 6.25 = 781.25 W
R2	100	125	125/100 = 1.25 A	125 × 1.25 = 156.25 W
R3	50	125	125/50 = 2.50 A	125 × 2.50 = 312.50 W
Total	12.5 Ω	125 V	10.00 A	1250.00 W

Part B — Schematic Component Tables

Circuit A — Series Circuit Component List

No.	Designator	Component	Value	Footprint	Qty
1	V1	DC Power Supply	125 V	—	1
2	R1	Resistor	20 Ω	R0603	1
3	R2	Resistor	30 Ω	R0603	1
4	R3	Resistor	50 Ω	R0603	1

Circuit B — Parallel Circuit Component List

No.	Designator	Component	Value	Footprint	Qty
1	V1	DC Power Supply	125 V	—	1
2	R1	Resistor	20 Ω	R0603	1
3	R2	Resistor	100 Ω	R0603	1
4	R3	Resistor	50 Ω	R0603	1

Arduino Nano + DHT22 Schematic Component List

No.	Designator	Component	Value / Comment	Footprint	Qty
1	U2	Arduino Nano	Microcontroller (5V, ATmega328P)	COMM-TH_ARDUINO_NANO	1
2	U3	DHT22 Sensor	Temperature & Humidity Sensor (3.3–5V)	SENSOR-TH_HAIGU	1
3	R1	Pull-up Resistor	10 kΩ	R0603	1

Wiring note: The DHT22 DATA pin is connected to Arduino Nano digital pin D2, with a 10 k Ω pull-up resistor between the DATA line and +5V. VDD is connected to the Nano's +5V pin, and GND to the Nano's GND.